

CHAPTER 16

BASICS OF BAYESIAN STATISTICS

16.1 INTRODUCTION

In the Basic Model introduced in Chapter 10, we pointed out that the process of making statistical decisions might benefit from additional information about the unknown parameter available prior to collecting the data. Such, usually subjective,³³ prior beliefs can then be updated with the (objective) information provided in the data using Bayes' theorem (4.4.1). Methods of statistical inference that use such prior information are called Bayesian Statistics.

As in methods of estimation presented in Chapter 11, observations $X_1, \dots, X_n$ are assumed to be a random sample from the distribution $f(x; \theta)$, and the objective is to estimate the value of θ . This time, however, we assume that θ is an element of some parameter space Θ , sampled according to probability distribution $\pi(\theta)$. For instance, if Θ is the real line, and Ξ is a continuous random variable with density $\pi(\theta)$, then $P\{\Xi \in A\} = \int_A \pi(\theta) d\theta$.^{*} The density $\pi(\theta)$ is referred to as the *prior* density of θ .

■ EXAMPLE 16.1

A grocery store receives shipments of some merchandise (e.g., oranges). The quality of each shipment can be described by a certain parameter θ .

Examples of such parameters may be the percentage of spoiled fruits, the average diameter of a fruit, the average sugar content of the juice, and so on. Typically, it is not practical to determine the exact value of θ for a given shipment, because of cost, or because sampling is destructive. In such cases, one takes a sample from a given shipment and uses it to estimate θ for this shipment.

If the store has been buying the shipments from the same company for some time, it usually has accumulated some data about the variability of values of θ from different

³³Subjective probability was discussed in Section 2.7.

shipments. Such historical data, in addition to the actual sample, could be used in assessment of the parameter θ for a given shipment.

■ EXAMPLE 16.2

In another example consider a physician who has to diagnose a patient. The data are various symptoms (or their lack), results of tests, and so on. The values of the parameter θ are different possible diagnoses, such as θ_1 = common cold, θ_2 = tuberculosis, and θ_3 = AIDS. The physician knows the distributions $f(\mathbf{x}, \theta)$ of the results $\mathbf{x} = (x_1, x_2, \dots, x_n)$ of n specific tests for every θ (here x_1 may be temperature, x_2 and x_3 may be systolic and diastolic blood pressure, x_4 may be glucose level, etc.). At some time, the physician reaches a diagnosis (i.e., reaches the decision, perhaps tentative, about θ and begins treatment). He may also order further tests. The estimation problems we have considered thus far have dealt only quantitative and usually continuous parameters. By contrasts, observe here how the discrete and qualitative nature of the parameter space affects the problem in the sense of a definition of error: while in estimation the error can be quantified as the difference $T(\mathbf{x}) - \theta$, in medical diagnosis such quantification is not possible (e.g., consider the “error” of treating a patient with TB as having a common cold, and vice versa, treating common cold as TB).

It is clear, however, that the experience and intuition of the diagnosing physician play a crucial role here. In particular, the physician may have some idea as to the prior probabilities $\pi(\theta)$ of various diseases.

16.2 PRIOR AND POSTERIOR DISTRIBUTIONS

The information from the sample, $\mathbf{x} = (x_1, \dots, x_n)$ (reflected in the likelihood function), and prior information about the distribution of θ (which quantifies what is known about the parameter before observing the sample), can be combined into the conditional density of θ given $\mathbf{x}$, referred to as the *posterior* density. We have

$$P\{\Xi \in A | \mathbf{X} = \mathbf{x}\} = \int_A \pi(\theta | \mathbf{x}) d\theta,$$

where

$$\pi(\theta | \mathbf{x}) = \frac{f_n(\mathbf{x}; \theta) \pi(\theta)}{\int_{\Theta} f_n(\mathbf{x}; u) \pi(u) du}. \quad (16.1)$$

The integral in the denominator of (16.1) is the reciprocal of the normalizing constant, and we often do not need to determine its value. What really matters is the fact that the posterior density of the parameter given the data is proportional to the likelihood of the data multiplied by the prior density (often referred to as a *kernel*):

$$\pi(\theta | \mathbf{x}) = C f_n(\mathbf{x}, \theta) \pi(\theta) = CL(\theta, \mathbf{x}) \pi(\theta) \propto L(\theta, \mathbf{x}) \pi(\theta).$$

The proportionality sign $\propto$ allows us to ignore factors that do not depend on the unknown parameter θ .

■ EXAMPLE 16.3

Suppose that we take observations from independent Bernoulli trials with the same probability of success θ , where θ follows a beta distribution with parameters α and β , that is,

$$\pi(\theta) = C\theta^{\alpha-1}(1-\theta)^{\beta-1}, \quad 0 \leq \theta \leq 1, \quad (16.2)$$

where C is the normalizing constant, given in (8.68). Letting $s = \sum x_i$, we have

$$f_n(\mathbf{x}, \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^s (1 - \theta)^{n-s}.$$

Consequently, the posterior density of θ , given $\mathbf{x}$, is

$$\pi(\theta|\mathbf{x}) \propto \theta^s (1 - \theta)^{n-s} \theta^{\alpha-1} (1 - \theta)^{\beta-1} = \theta^{\alpha+s-1} (1 - \theta)^{n+\beta-s-1},$$

which we recognize as a $\text{BETA}(\alpha + s, \beta + n - s)$ distribution.

■ EXAMPLE 16.4

Assume now that the observations $X_1, X_2, \dots, X_n$ are independent, each with $\text{POI}(\theta)$ distribution, while θ has a $\text{GAM}(\alpha, \lambda)$ distribution so that

$$\pi(\theta) = C \theta^{\alpha-1} e^{-\lambda\theta}, \quad \theta > 0, \quad (16.3)$$

where C is the normalizing constant given in (8.49). Letting $s = \sum x_i$, the posterior density of θ is

$$\pi(\theta|\mathbf{x}) = \left[\prod_{i=1}^n \left(\frac{\theta^{x_i}}{x_i!} e^{-\theta} \right) \right] C \theta^{\alpha-1} e^{-\lambda\theta} = K \theta^{s+\alpha-1} e^{-(n+\lambda)\theta}. \quad (16.4)$$

Here the constant K also incorporates factorials $x_1! \cdots x_n!$ (in general, we only need to keep track of the terms involving θ). We recognize (16.4) as a density of a $\text{GAM}(\alpha + s, \lambda + n)$ distribution.

■ EXAMPLE 16.5

Suppose that $X_1, \dots, X_n$ is a random sample from a $\text{N}(\theta, \sigma^2)$ distribution with σ^2 known and that θ has a prior normal distribution $\text{N}(\mu, \tau^2)$. We will find the posterior distribution of θ , given the sample.

SOLUTION. The likelihood function can be transformed as follows:

$$\begin{aligned} f_n(\mathbf{x}, \theta) &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum (x_i - \theta)^2 \right\} \\ &\propto \exp \left\{ -\frac{1}{2\sigma^2} \left[\sum (x_i - \bar{x}_n)^2 + n(\theta - \bar{x}_n)^2 \right] \right\} \\ &\propto \exp \left\{ -\frac{n}{2\sigma^2} (\theta - \bar{x}_n)^2 \right\}. \end{aligned}$$

The sign $\propto$ means proportionality. The ignored factors are $\sigma^{-n} (2\pi)^{-n/2}$ in the first proportionality, and $\exp\{-(n/2\sigma^2)(x_i - \bar{x}_n)^2\}$ in the second proportionality. The posterior density $\pi(\theta|\mathbf{x})$ is therefore proportional to

$$\exp \left\{ -\frac{n}{2\sigma^2} (\theta - \bar{x}_n)^2 \right\} \times \exp \left\{ -\frac{1}{2\tau^2} (\theta - \mu)^2 \right\}.$$

Now we need to separate the terms involving θ ; all other terms will be absorbed in the proportionality constant. Expanding the squares and leaving only terms with θ , we obtain, after some algebra,

$$\pi(\theta|\mathbf{x}) \propto \exp \left\{ -\frac{1}{2q^2} (\theta - m)^2 \right\},$$

where

$$m = \frac{\sigma^2}{\sigma^2 + n\tau^2}\mu + \frac{n\tau^2}{\sigma^2 + n\tau^2}\bar{x}_n \quad (16.5)$$

and

$$q^2 = \frac{\sigma^2\tau^2}{\sigma^2 + n\tau^2}. \quad (16.6)$$

This means that the posterior density of θ is again normal with mean m and variance q^2 given by (16.5) and (16.6), respectively.

■ EXAMPLE 16.6

Suppose finally that the observations $X_1, \dots, X_n$ are independent normal $N(0, \theta^2)$, with θ again having a gamma distribution as in Example 16.4. We have

$$\pi(\theta|\mathbf{x}) = K \left(\prod_{i=1}^n \frac{1}{\theta\sqrt{2\pi}} e^{-\frac{x_i^2}{2\theta^2}} \right) \theta^{\alpha-1} e^{-\lambda\theta} = K^* \theta^{\alpha-n-1} e^{-\lambda\theta - \frac{\sum x_i^2}{2\theta^2}}. \quad (16.7)$$

This time the density (16.7) is not a member of any known family of distributions.

The following definition is related to the situation in Examples 16.3 through 16.5 but not in Example 16.6:

Definition 16.2.1 A family $\mathcal{F}$ of distributions of parameter θ is said to provide *conjugate* priors for the distribution $f(x, \theta)$ of observations if, whenever the prior distribution of θ is in $\mathcal{F}$, the posterior distribution $\pi(\theta|\mathbf{x})$ also belongs to $\mathcal{F}$ for any sample $\mathbf{x}$. $\square$

The essence of Examples 16.3 through 16.5 is formulated in the following theorems:

Theorem 16.2.1 *Beta densities are conjugate priors for the binomial distribution.*

Theorem 16.2.2 *Gamma densities are conjugate priors for the Poisson distribution.*

Theorem 16.2.3 *Normal densities are conjugate priors for normal distribution.*

Conjugate priors simplify the process of obtaining posterior distributions. Rather than computing integrals, one only needs to modify parameters of the prior distribution given the data, as will be illustrated in Examples 16.14–16.16.

In the following example, both distributions will be discrete.

■ EXAMPLE 16.7

Let us assume that random variable X attains four different values: a, b, c, d with the following probabilities:

X	a	b	c	d
$P(X; \theta)$	θ^2	$\theta(1 - \theta)$	$\theta(1 - \theta)$	$(1 - \theta)^2$

and that the prior distribution of θ is also discrete:

θ	1/4	1/2	3/4
$\pi(\theta)$	1/2	1/3	1/6

Suppose now that value a of variable X was observed. Since

$$\begin{aligned}
 P(a) &= P(a; \theta = 1/4)\pi(1/4) + P(a; \theta = 1/2)\pi(1/2) \\
 &\quad + P(a; \theta = 1/3)\pi(1/3) \\
 &= \left(\frac{1}{4}\right)^2 \times \frac{1}{2} + \left(\frac{1}{2}\right)^2 \times \frac{1}{3} + \left(\frac{3}{4}\right)^2 \times \frac{1}{6} = \frac{10}{48},
 \end{aligned}$$

we have

$$\begin{aligned}
 \pi(\theta = 1/4|a) &= \left(\frac{1}{4}\right)^2 \times \frac{1}{4} / \left(\frac{10}{48}\right) = \frac{3}{20}, \\
 \pi(\theta = 1/2|a) &= \left(\frac{1}{2}\right)^2 \times \frac{1}{2} / \left(\frac{10}{48}\right) = \frac{8}{20}, \\
 \pi(\theta = 3/4|a) &= \left(\frac{3}{4}\right)^2 \times \frac{1}{6} / \left(\frac{10}{48}\right) = \frac{9}{20},
 \end{aligned}$$

so that the posterior distribution $\pi(\theta|a)$ is

θ	1/4	1/2	3/4
$\pi(\theta a)$	3/20	8/20	9/20

Let us now assume that an additional value was sampled and it occurred to be a d . Using posterior distribution $\pi(\theta|a)$ as a prior we now have

$$\begin{aligned}
 P(d) &= \left(1 - \frac{1}{4}\right)^2 \times \frac{6}{20} + \left(1 - \frac{1}{2}\right)^2 \times \frac{8}{20} \\
 &\quad + \left(1 - \frac{3}{4}\right)^2 \times \frac{9}{20} = \frac{17}{80}
 \end{aligned}$$

and the posterior distribution $\pi(\theta|a, d)$ becomes

θ	1/4	1/2	3/4
$\pi(\theta a, d)$	27/68	32/68	9/68

This example shows how the information is updated in a natural way. This sequential nature of Bayesian theorem is called recursive Bayesian learning and is the basis for all Bayesian modeling.

The prior distribution represents the current knowledge or the description of uncertainty about the model parameters. There are different types of prior distributions. *Informative priors* are based on either historical data or they reflect the degree of belief about the unknown parameters. If these priors have a large impact on posterior distributions, we say that they

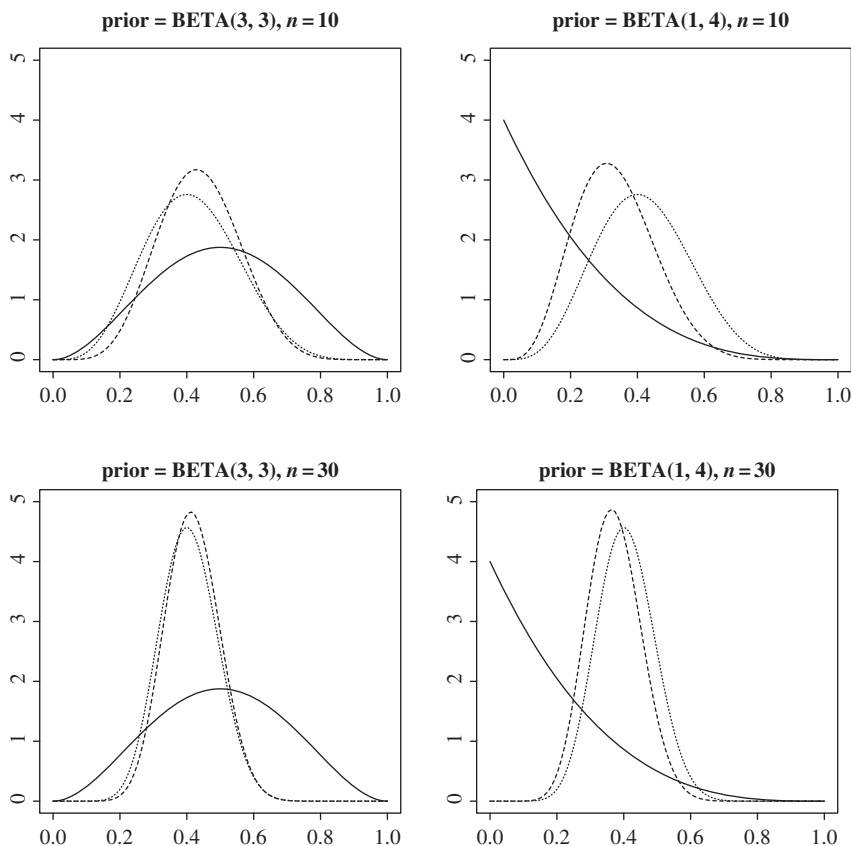


Figure 16.1 Posterior densities (dashed line) for different prior densities (solid line) and different likelihood (dotted line).

dominate the likelihood. Some such priors can be seen in the top line of Figure 16.1 and in Examples 16.3–16.6.

Noninformative priors (also known as *reference*, *vague*, or *objective*) usually represent the lack of information about parameters in the model. Their impact on posterior distributions is limited as they assign equal probability (density) to all values of the parameter within a certain range (also infinite). However, the name “noninformative” is not always the best as these priors do not necessarily represent ignorance. Sometimes, for example, they support extreme values of the parameter.

■ EXAMPLE 16.8

In Example 16.4, prior and posterior distributions of the probability of success θ were $\text{BETA}(\alpha, \beta)$ and $\text{BETA}(\alpha + s, \beta + n - s)$, respectively. For $\alpha = \beta = 1$, the prior distribution $\text{BETA}(1, 1)$ would have a constant density $\pi(\theta) = 1$ for $0 \leq \theta \leq 1$ and will become a noninformative prior.

While $\text{BETA}(1, 1)$ is a well defined proper distribution, some noninformative distributions are *improper* priors.

Definition 16.2.2 A prior $\pi(\theta)$ is said to be improper if $\int_{\Theta} \pi(\theta) d\theta = \infty$. □

An improper prior may lead to an improper posterior and no inference can be made in such a case. In other situations, improper priors can lead to proper posteriors, as will be illustrated in the following examples.

■ **EXAMPLE 16.9**

Let us assume that $X_1, \dots, X_n$ is a random sample from a $N(\theta, 1)$ distribution and that $-\infty < \theta < \infty$. Moreover, let $\pi(\theta) = 1$, which is an improper prior. The posterior density $\pi(\theta|\mathbf{x})$ is then

$$\begin{aligned}\pi(\theta|\mathbf{x}) &\propto \pi(\mathbf{x}|\theta)\pi(\theta) \\ &\propto \exp\left\{-\frac{1}{2}\sum (x_i - \theta)^2\right\} \times 1 \\ &\propto \exp\left\{-\frac{1}{2}(n\theta^2 - 2\theta\sum x_i)\right\} \\ &\propto \exp\left\{-\frac{n}{2}(\theta - \bar{x})^2\right\}.\end{aligned}$$

The improper prior lead to a proper posterior distribution $N(\bar{x}, 1/n)$.

■ **EXAMPLE 16.10**

Let us now have a random sample $X_1, \dots, X_n$ from a $\text{POI}(\theta)$ distribution and an improper prior $\pi(\theta) = \theta^{-1}$ for $0 < \theta < \infty$. Letting $s = \sum x_i$, the posterior density $\pi(\theta|\mathbf{x})$ is then proportional to

$$\begin{aligned}\pi(\mathbf{x}|\theta)\pi(\theta) &\propto \left(\prod \frac{e^{-\theta}\theta^{x_i}}{x_i!}\right) \times \frac{1}{\theta} = \frac{e^{-n\theta}\theta^s \times \theta^{-1}}{\prod x_i!} \\ &\propto e^{-n\theta} \times \theta^{s-1},\end{aligned}$$

which is a kernel of the $\text{GAM}(\sum x_i, n)$ distribution, a proper posterior.

The most often used noninformative (or reference) prior is the so-called Jeffreys prior.

Definition 16.2.3 Jeffreys prior $\pi(\theta)$ is proportional to the square root of the Fisher information $I(\theta)$:

$$\pi(\theta) \propto |I(\theta)|^{1/2}. \quad \square$$

Jeffreys priors can be either proper or improper as will be shown in the following examples.

■ **EXAMPLE 16.11**

Let $X_1, \dots, X_n$ be a random sample from a Bernoulli distribution $\text{BIN}(1, \theta)$. Since $I(\theta) = [\theta(1 - \theta)]^{-1}$ (see Example 11.16), Jeffreys prior is

$$\pi(\theta) \propto \left(\frac{1}{\theta(1 - \theta)}\right)^{1/2} = \theta^{-1/2}(1 - \theta)^{-1/2},$$

which is $\text{BETA}(1/2, 1/2)$, a proper prior.

In the next example, Jeffreys prior is improper.

■ **EXAMPLE 16.12**

Let $X_1, \dots, X_n$ be a random sample from a POI(θ) distribution. The Fisher information is now

$$I(\theta) = \theta^{-1}$$

and the Jeffreys prior $\pi(\theta) = \theta^{-1/2}$ is improper since

$$\int_0^\infty \pi(\theta) d\theta = \int_0^\infty \theta^{-1/2} d\theta = \infty.$$

An advantage of using a Jeffreys prior is that it is invariant under different parametrizations of θ . If h is a one-to-one function of θ then

$$\begin{aligned} I(h(\theta)) &= -E \left(\frac{d^2 \log \pi(x|h)}{dh^2} \right) \\ &= -E \left(\frac{d^2 \log \pi(x|h)}{d\theta^2} \times \left| \frac{d\theta}{dh} \right|^2 \right) \\ &= I(\theta) \times \left| \frac{d\theta}{dh} \right|^2 \end{aligned} \tag{16.8}$$

and

$$I(h(\theta))^{1/2} = I(\theta)^{1/2} \left| \frac{d\theta}{dh} \right| = \pi(\theta) \left| \frac{d\theta}{dh} \right| = \pi(h(\theta)).$$

If we are unsure which prior distribution should be used, then we can consider different possible priors and compare the resulting posterior distributions. If the data provide a lot more information than the prior (the prior is dominated by the likelihood), then the posterior distributions will be very similar with different priors, an example may be seen in the lower part of Figure 16.1. If there are multiple parameters in the model, assigned prior distributions (or probabilities) are usually different as they need to be relevant for different parameters.³⁴

PROBLEMS

- 16.2.1** Six randomly selected adults are asked if they favor additional taxes to help fund more affordable health care. Four of them respond “yes.” Assuming that the prior distribution is BETA(1, 1), determine the posterior density and obtain the probability that in the population the proportion of people favoring a tax increase is between 60% and 70%.
- 16.2.2** Determine posterior distributions to show that the family of gamma distributions provides conjugate priors for: **(i)** The exponential distribution. **(ii)** The BETA (α, β) distribution with β known.

³⁴More information about noninformative priors can be found in the literature, see, e.g., Kass and Wasserman (1989).

16.2.3 If random variable X has a $\text{GAM}(\alpha, \lambda)$ distribution with density

$$f(x; \alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$$

for $x > 0$, then variable $Y = 1/X$ has a so-called *inverse gamma* distribution, $\text{IGAM}(\alpha, \lambda)$, with density

$$f(y; \alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{-\alpha-1} e^{-\lambda/y}, \quad (16.9)$$

for $y > 0$, with $E(Y) = \lambda/(\alpha - 1)$ for $\alpha > 1$ and $\text{Var}(Y) = \lambda^2/[(\alpha - 1)^2(\alpha - 2)]^{-1}$ for $\alpha > 2$. The mode of $\text{IGAM}(\alpha, \lambda)$ is $\frac{\lambda}{\alpha+1}$. Show that the inverse gamma distribution provides a conjugate prior for parameter σ^2 in $N(\mu, \sigma^2)$ when μ is known.

16.3 BAYESIAN INFERENCE

Most inferential questions can be answered by appropriate analysis of the posterior distribution. Such analysis can provide point and interval estimates of parameters, predict distributions of future observations, and also compare hypotheses.

Predictive Distribution

Let us now assume that we have the data $\mathbf{x}$, decided on a prior distribution $\pi(\theta)$, and would like to obtain $f(y|\mathbf{x})$ —a *Bayesian predictive distribution* of the future observation y . Since the random sample $\mathbf{X}$ and Y , the future observation from the same population, are conditionally independent given the value of θ , we have

$$\begin{aligned} f(y|\mathbf{x}) &= \frac{f_{n+1}(y, \mathbf{x})}{f_n(\mathbf{x})} = \frac{\int f_{n+1}(y, \mathbf{x}; \theta) \pi(\theta) d\theta}{\int f_n(\mathbf{x}; \theta) \pi(\theta) d\theta} \\ &= \frac{\int f(y, \theta) f_n(\mathbf{x}; \theta) \pi(\theta) d\theta}{\int f_n(\mathbf{x}; \theta) \pi(\theta) d\theta} \\ &= \int f(y; \theta) \times \frac{f_n(\mathbf{x}; \theta) \pi(\theta)}{\int f_n(\mathbf{x}; \theta) \pi(\theta) d\theta} d\theta \\ &= \int f(y; \theta) \pi(\theta|\mathbf{x}) d\theta. \end{aligned} \quad (16.10)$$

■ EXAMPLE 16.13

Suppose that $X_1, \dots, X_n$ is a random sample from a $\text{POI}(\theta)$ distribution and that a prior distribution of θ is $\text{GAM}(\alpha, \lambda)$. Using this information, we will predict a distribution of an additional observation y taken from the same population. Based on Example 16.4, the posterior distribution $\pi(\theta|\mathbf{x})$ is $\text{GAM}(\sum x_i + \alpha, \lambda + n)$ with the density

$$\frac{(\lambda + n)^{\sum x_i + \alpha}}{\Gamma(\sum x_i + \alpha)} \theta^{\sum x_i + \alpha - 1} e^{-(\lambda + n)\theta}.$$

Based on (16.10), the density of the predictive distribution is

$$\begin{aligned} f(y|\mathbf{x}) &= \int_0^\infty \frac{\theta^y e^{-\theta}}{y!} \times \frac{(\lambda + n)^{\sum x_i + \alpha}}{\Gamma(\sum x_i + \alpha)} \theta^{\sum x_i + \alpha - 1} e^{-(\lambda + n)\theta} d\theta \\ &= \frac{(\lambda + n)^{\sum x_i + \alpha}}{y! \Gamma(\sum x_i + \alpha)} \int_0^\infty \theta^{y + \sum x_i + \alpha - 1} e^{-\theta(\lambda + n + 1)} d\theta. \end{aligned} \quad (16.11)$$

Since the integrand is the kernel of the $\text{GAM}(y + \sum x_i + \alpha, \lambda + n + 1)$ distribution, the integral equals

$$\frac{\Gamma(y + \sum x_i + \alpha)}{(\lambda + n + 1)^{y + \sum x_i + \alpha}}$$

and the density of the predictive distribution is

$$\begin{aligned} f(y|\mathbf{x}) &= \frac{(\lambda + n)^{\sum x_i + \alpha}}{y! \Gamma(\sum x_i + \alpha)} \times \frac{\Gamma(y + \sum x_i + \alpha)}{(\lambda + n + 1)^{y + \sum x_i + \alpha}} \\ &= \binom{y + \sum x_i + \alpha - 1}{y} \left(\frac{\lambda + n}{\lambda + n + 1} \right)^{\sum x_i + \alpha} \left(\frac{1}{\lambda + n + 1} \right)^y. \end{aligned} \quad (16.12)$$

Thus, the predictive distribution is $\text{NBIN}(\sum x_i + \alpha, 1/(\lambda + n + 1))$.

Since the choice of the prior distribution is subjective, it is to be expected that two different people might want to use different prior distributions. Therefore, it is worthwhile to conduct a *sensitivity analysis*—compute posterior distributions for different possible priors and see how different they are, i.e., how strongly the different prior opinions affect the posterior (see Figure 16.1). Sensitivity in the choice of prior is not necessarily a disadvantage. It indicates that the data do not provide sufficient information and that perhaps additional observations should be collected.

Point Estimation

The choice of the estimator depends on the choice of the loss function. In general, as in Section 11.3, the penalty for accepting the value of the parameter as θ^* (while in fact it is θ) is expressed by the loss function $\mathcal{L}(\theta^*, \theta)$. In the present case, we know the distribution of θ , given the observed sample $\mathbf{x}$. Thus, we should choose the value θ^* that minimizes the expected loss

$$E\{\mathcal{L}(\theta^*, \Xi)|\mathbf{x}\} = \int_{\Theta} \mathcal{L}(\theta^*, \theta) \pi(\theta|\mathbf{x}) d\theta. \quad (16.13)$$

The left-hand side of (16.13) *does not depend on* θ . It is a function of θ^* and $\mathbf{x}$. For each $\mathbf{x}$, we can therefore try to minimize the expectation, that is, find a value $\theta^*(\mathbf{x})$ such that

$$E\{\mathcal{L}(\theta^*(\mathbf{x}), \Xi)|\mathbf{x}\} \leq E\{\mathcal{L}(\theta^{**}, \Xi)|\mathbf{x}\}$$

for every θ^{**} in Θ . Such a value $\theta^*(\mathbf{x})$ is the best choice of estimate given the sample $\mathbf{x}$. When $\mathbf{x}$ varies according to the distribution $f(\mathbf{x}, \theta)$, we obtain a statistic $\theta^*(\mathbf{X})$. We will now introduce the following definition:

Definition 16.3.1 The statistic $\theta^*(X)$ that minimizes the left-hand side of (16.13) is called the *Bayes estimator of θ for the loss function $\mathcal{L}(\theta^*, \theta)$* . $\square$

Minimizing (16.13) in the general case may not be easy. However, the minimizing value θ^* is well known when $\mathcal{L}(\theta^*, \theta) = (\theta^* - \theta)^2$. It is the mean of the posterior distribution of θ given $\mathbf{x}$. As in the case of other estimators, we will tacitly take the squared error as the loss function (unless we explicitly specify some other loss function). This is the customary choice. It is motivated primarily by the fact that the quadratic loss function allows further development of the theory. Therefore, if the loss function is not explicitly specified as other than quadratic, then the Bayes estimator of parameter θ is understood as the mean of the posterior distribution:

$$T(\mathbf{X}) = E\{\Xi|\mathbf{X}\} = \int \theta \pi(\theta|\mathbf{X}) d\theta. \quad (16.14)$$

The following theorem is analogous to the Theorem 11.6.2:

Theorem 16.3.1 *The Bayes estimator of θ is a function of the minimal sufficient set of statistics for θ .*

Proof: If statistics $T_1, \dots, T_k$ are jointly sufficient for θ , then the posterior density of θ given $\mathbf{x}$ is, for prior density $\pi(\theta)$, is

$$\begin{aligned} \pi(\theta|\mathbf{x}) &= \frac{f_n(\mathbf{x}, \theta)\pi(\theta)}{\int f_n(\mathbf{x}, \eta)\pi(\eta)d\eta} = \frac{v(\mathbf{x})u(T_1, \dots, T_k, \theta)\pi(\theta)}{\int v(\mathbf{x})u(T_1, \dots, T_k, \eta)\pi(\eta)d\eta} \\ &= \frac{u(T_1, \dots, T_k, \theta)\pi(\theta)}{H(T_1, \dots, T_k)}. \end{aligned}$$

Since the factor $v(\mathbf{x})$ cancels, H is a function of statistics $T_1, \dots, T_k$. Consequently, the Bayes estimator, equal to the mean of posterior distribution, depends on $\mathbf{x}$ only through statistics $T_1, \dots, T_k$. $\square$

■ EXAMPLE 16.14

If the observations $\mathbf{X} = (X_1, \dots, X_n)$ form a random sample from a Bernoulli distribution (so that $S = \sum_{i=1}^n X_i$ is $\text{BIN}(n, \theta)$), and θ varies according to $\text{BETA}(\alpha, \beta)$, then the Bayes estimator of θ is

$$T(X_1, \dots, X_n) = \frac{\alpha + S}{\alpha + \beta + n}. \quad (16.15)$$

This follows from the fact, established in Example 16.3, that the posterior distribution of θ is $\text{BETA}(\alpha + S, \beta + n - S)$ with the mean $(\alpha + S)/(\alpha + \beta + n)$.

Since

$$\frac{\alpha + S}{\alpha + \beta + n} = \frac{\alpha + \beta}{\alpha + \beta + n} \times \frac{\alpha}{\alpha + \beta} + \frac{n}{\alpha + \beta + n} \times \frac{S}{n}, \quad (16.16)$$

where $\alpha/(\alpha + \beta)$ is the expectation of the prior distribution and S/n is the sample mean, the posterior expectation can be interpreted as a weighted average of the prior expectation and the sample average.

■ EXAMPLE 16.15

In a similar way, we can use Theorem 11.4.1, which asserts that gamma densities are conjugate priors for the Poisson distribution. If $X_1, \dots, X_n$ is a sample from $\text{POI}(\theta)$ distribution, and θ has prior distribution $\text{GAM}(\alpha, \lambda)$, then the Bayes estimator of θ is

$$T(X_1, \dots, X_n) = \frac{\alpha + S}{\lambda + n}. \quad (16.17)$$

This follows from the fact that the mean of a gamma distribution is the ratio of the shape parameter to the scale parameter. As in Example 16.14, the Bayes estimator can again be expressed as a weighted average of the prior expectation α/λ and the sample mean S/n :

$$\frac{\alpha + S}{\lambda + n} = \frac{\lambda}{\lambda + n} \times \frac{\alpha}{\lambda} + \frac{n}{\lambda + n} \times \frac{S}{n}.$$

■ EXAMPLE 16.16

From (16.5) in Example 16.5 it follows that the Bayes estimator in the case of the normal distribution $N(\theta, \sigma^2)$, with known σ^2 and normal prior $N(\mu, \tau^2)$, is

$$T = \frac{\sigma^2}{\sigma^2 + n\tau^2} \mu + \frac{n\tau^2}{\sigma^2 + n\tau^2} \bar{X}_n. \quad (16.18)$$

Thus T , as in Examples 16.14 and 16.15, is a weighted average of prior mean μ and sample mean $\bar{X}_n$. With the increase of the sample size n , the estimator puts more and more weight to the sample size $\bar{X}_n$, and in the limit we have $T = \bar{X}_n$. In other words, ultimately the empirical evidence always prevails over prior convictions. Finally, the same limiting conclusion is obtained if $\tau^2 \rightarrow \infty$, that is, when the prior information is more vague (the prior variance τ^2 is interpretable as a measure of uncertainty of the prior information). Going toward the opposite extreme, where $\sigma^2 \rightarrow \infty$ or $\tau^2 \rightarrow 0$ (observations are subject to large errors, or prior knowledge has high certainty), the Bayes estimator of θ attaches more and more weight to the prior information that the mean of θ is μ .

The last three examples illustrate the convenience of using conjugate prior distributions. This convenience lies basically in the fact that we have a simple formula for the Bayes estimator, and therefore we can easily adjust our estimates when new observations become available. To illustrate this point, we consider the following example:

■ EXAMPLE 16.17

Continuing Example 16.15, assume that we first take m observations, with the total number of successes $S = X_1 + \cdots + X_m$, and then n observations, with the total number of successes $S' = X'_1 + \cdots + X'_n$. The prior density of θ , before taking the first set of observations, is $\text{BETA}(\alpha, \beta)$.

The situation at the end of second series of observations can be regarded in two ways:

1. We have a total of $S + S'$ successes in $m + n$ trials, so the Bayes estimator is

$$\frac{\alpha + (S + S')}{\alpha + \beta + (m + n)}. \quad (16.19)$$

2. We have a total of S' successes in the last n trials, where the new prior distribution becomes a posterior distribution after we take the first m observations. The posterior density is $\text{BETA}(\alpha_1, \beta_1)$, where $\alpha_1 = \alpha + S$ and $\beta_1 = \beta + m - S$. The Bayes estimator is now

$$\frac{\alpha_1 + S'}{\alpha_1 + \beta_1 + n},$$

the same as (16.19).

Essentially what we observed here is an instance of a general theorem on updating evidence (Theorem 4.4.2), which says that if we have two independent (sets of) observations $\mathbf{x}$ and $\mathbf{x}'$, we can use them either “at once,” to determine the posterior distribution of θ given $(\mathbf{x}, \mathbf{x}')$, or we can do it in steps. That is, we find the posterior density of θ , given one data set (e.g., $\mathbf{x}$), and we use it as a new prior to find the posterior density of θ given $\mathbf{x}'$. The results will still be the same. Moreover, the order of choice between $\mathbf{x}$ and $\mathbf{x}'$ is irrelevant, and does not need to coincide with the order in which the data $\mathbf{x}$ and $\mathbf{x}'$ were collected.

All the examples above concern estimators for the squared error loss function. If the loss function is $\mathcal{L}(\theta^*, \theta) = |\theta^* - \theta|$, then the Bayes estimator is the median of the posterior distribution (see Theorem 7.6.3). For the normal case (Example 16.16), the mean and median coincide; hence, (16.18) is also the Bayes estimator for the absolute error loss. In Examples 16.14 and 16.15, the posterior distributions are beta or gamma, and their medians are not expressible by simple formulas in terms of the parameters.

Loss functions other than the squared or absolute error are also used. Asymmetric loss functions (introduced in Chapter 11) are often of importance in data mining procedures.

It is clear that the class of *all* distributions on the parameter space Θ is always a set of conjugate priors for any distributions. Such a statement, however, is totally pointless. In fact, a class of conjugate priors is useful only if it leads to a simple formula for the posterior density. This allows explicit formulas for the means and thus for Bayesian estimators against mean squared loss. From this perspective, the fact that, a given class of distributions is a class of conjugate priors is just a mathematical curiosity without much significance. Indeed, referring again to Example 16.17, we consider there a situation where observations $S = \sum X_i$ are binomial with parameter θ , while θ has a beta distribution. The first assumption is defensible: we often can make S have a binomial distribution by using an appropriate sampling scheme. But the law that governs the variability of θ from case to case is beyond our control, and the class of situations described above, where the parameter θ is a value of a random variable Ξ with distribution π , is rather restricted. Most often, the situation that a statistician faces is “one of a kind,” characterized by an unknown value of θ . As such, it does not make sense to think of a prior distribution π of θ as telling us “how often” we had analogous statistical problems in which the value of the parameter satisfied the inequality of the form $a \leq \theta \leq b$.

There is a view that, even in such “one of a kind” situations, it makes sense to consider and use the prior distribution of the parameter. Actually, the issue of using prior distributions—even if they do not represent frequencies of occurrence of situations characterized by some values of θ —divides statisticians into Bayesians and non-Bayesians.

The philosophical points of this division are beyond the scope of this book. One could, however, consider the following two competing principles.

1. Statistical conclusions should depend on data only. When two statisticians analyze the same data using the same method, they should reach the same conclusion.
2. Statistical conclusions may depend on the experience, intuition, and insight of the statistician who analyzes a given set of data.

Very roughly, statisticians who adhere strictly to principle 1 are non-Bayesians, and those who favor 2 are Bayesians. The latter use the prior distribution π as a means to express their knowledge, intuition, and so forth. In this respect, having a class of conjugate priors is usually of great help, primarily to express one’s own prior experience or convictions, or to elicit information about the analyzed problem from the practitioners whom they advise. However, the first example below shows something more fundamental, namely that the concept of prior distribution, reflecting one’s personal judgments about a “one-of-a-kind” case, is sometimes unavoidable. Consider the following situation, which without this concept appears paradoxical:

■ EXAMPLE 16.18

Imagine yourself playing the following game: There are two envelopes, each containing a check. The amount on one check is twice as big as the amount on the other. You choose an envelope and inspect the check. At this moment you are offered an option to choose the other envelope. What should you do?

SOLUTION. The standard reasoning is as follows: Let a be the amount on the check that was in the first envelope you selected. The other envelope then contains a check for either $2a$ or $a/2$, each with the same probability. Thus, if you change your decision, the expected outcome is

$$\frac{1}{2}(2a) + \frac{1}{2}\left(\frac{a}{2}\right) = \frac{5}{4}a > a, \quad (16.20)$$

which means that *you should always change* the envelope.

This may seem paradoxical, since money appears to be created out of nowhere, just by changing the decision. The explanation lies in the fact that calculation (16.20) of the expected value uses probabilities 0.5 that the other envelope contains checks for $2a$ or $a/2$. In fact, one should use here the conditional probabilities, given the observed value a , and these calculations involve prior probabilities.

Indeed, suppose that $\pi(x)$ is the prior probability that the envelopes contain checks for the amounts x and $2x$. Assuming that one has no clairvoyant abilities, and therefore always has the chance 0.5 of selecting an envelope with the lesser amount on a check, the unconditional probability of observing the amount a is $0.5\pi(a/2) + 0.5\pi(a)$. Given the observed amount a , the probability that the check in the other envelope is for the amount $a/2$ equals

$$\frac{0.5\pi(a/2)}{0.5\pi(a/2) + 0.5\pi(a)} = \frac{\pi(a/2)}{\pi(a/2) + \pi(a)}.$$

The analogous probability for the amount $2a$ is

$$\frac{0.5\pi(a)}{0.5\pi(a/2) + 0.5\pi(a)} = \frac{\pi(a)}{\pi(a/2) + \pi(a)}.$$

But the condition

$$\frac{\pi(a/2)}{\pi(a/2) + \pi(a)} = \frac{\pi(a)}{\pi(a/2) + \pi(a)}$$

implies $\pi(a/2) = \pi(a)$, which cannot be satisfied for all a (regardless of whether π represents a density or a discrete probability distribution).

This argument refers to some prior distribution π on the possible amount on the lesser check. Whether this distribution has any frequential interpretation (referring to analogous games played before), or not (if such a game is played only once) is irrelevant here. The only way to escape the paradox is to realize that everyone has some idea as to the probable range of values that may appear on the checks in the game. If the check in hand shows a very “small” value, then the other is probably for a higher value. If the check in hand shows a very “high” value, the other is probably smaller. The concepts “very small,” “very large,” and “probably” are subjective here and refer to the player’s idea about the distribution π .

This example provides one more (and rather powerful) argument for the need of the Bayesian approach to statistical problems.

■ EXAMPLE 16.19

A piece of rock (e.g., taken from the moon) is sent to a laboratory to determine its radioactivity level. Assume that the measurement is simply a Geiger count N_t , recorded for certain time t . The role of the number of observations n is now played by observation time t . To avoid confusion, let us assume simply that the experiment is run in such a way that Geiger counts $X_1, X_2, \dots$ are recorded, where X_i is the total count of radioactive particles in the i th hour of observation. Then, $N_t = X_1 + \dots + X_t$ if t is an integer number of hours.

We know that N_t has a Poisson distribution with mean θt , where θ is the radiation intensity expressed in average number of emissions per hour. Suppose that the initial estimate of θ is needed urgently, so that observations can be carried out only for a limited time T . In other words, we have at our disposal a single observation of N_t . The likelihood function here is

$$L(\theta) = \frac{(\theta T)^{N_T}}{(N_T)!} e^{-\theta T},$$

therefore

$$\log L(\theta) = C + N_T \log \theta - \theta T,$$

and the MLE is easily seen to be

$$\hat{\theta} = \frac{N_T}{T}$$

(note that this result concerns the MLE in the case of observations running continuously in time; we no longer have a sample of size n). For example, assume that a total of 50 counts was recorded in $T = 100$ hours of observations so that the MLE of θ is 0.5.

Suppose now that there are two physicists in the laboratory, and each has his or her own ideas about what the radioactivity level θ of the specimen tested might be. Dr. Brown favors a certain theory of how the moon was formed, and how and when its rocks became initially radioactive. She thinks that moon rocks should have their level of radioactivity θ about 1, but she is willing to incorporate a fair amount of uncertainty in her judgment, allowing the standard deviation of θ to be as much as 50% of the mean.

On the contrary, Dr. Smith favors a theory which predicts that the moon should have uniformly low radioactivity, say $\theta = 0.4$ on average, with standard deviation not exceeding 5% of the mean.

Let us see how these prior convictions will affect the estimates of θ for the specimen in question. We assume that the prior densities belong to the gamma family. If the parameters of a gamma distribution are α (shape) and λ (scale), then the mean is α/λ , and the variance is α/λ^2 ; see (8.50) and (8.51). Consequently, the ratio of the standard deviation to the mean (the so-called coefficient of variation) is

$$CV = \frac{\sqrt{\alpha/\lambda^2}}{\alpha/\lambda} = \frac{1}{\sqrt{\alpha}}.$$

Thus, for Dr. Brown, we have

$$CV = \frac{1}{\sqrt{\alpha}} = 0.5, \quad \frac{\alpha}{\lambda} = 1,$$

which gives $\alpha = \lambda = 4$.

For Dr. Smith, we have

$$CV = \frac{1}{\sqrt{\alpha}} = 0.05, \quad \frac{\alpha}{\lambda} = 0.4,$$

so $\alpha = 400$, $\lambda = 1,000$.

The Bayes estimator is (see Example 16.15) $(\alpha + N_T)/(\lambda + T)$, so that

$$\hat{\theta}_B = \frac{4 + 50}{4 + 100} = 0.519 \quad \text{and} \quad \hat{\theta}_S = \frac{400 + 50}{1,000 + 100} = 0.409$$

for Dr. Brown and Dr. Smith, respectively.

We can see the effects of two factors. One is that the MLE (in this case equal 0.5) is being “pulled” toward the mean of the prior distribution. The amount of pull depends on the variance of the prior distribution, reflecting the strength of conviction in the prior distribution: Dr. Smith, whose prior has much smaller variance, ends up with an estimate much closer to his prior mean than does Dr. Brown.

The next example concerns the behavior of certain species of birds in their search for food. The complete theory, including optimization aspects, has been developed by Green (1980). We present here only a fragment concerning assessment (estimation) by a bird.

■ EXAMPLE 16.20 Are Birds Bayesians?

Assume that the species in question can find food only in “patches,” each consisting of a certain number of places where prey can be found. Specifically, we will think of birds that prey on worms that live in pine cones. We assume that a pine cone has n “holes,” each of them containing a prey with probability θ , independently of other holes. Thus, given θ , the number of prey in a cone has a $\text{BIN}(n, \theta)$ distribution. The bird has a fixed search pattern, so it does not search the same hole twice, and we assume that the prey cannot hide or escape to another hole during the search. Finally, we assume that θ , the frequency of occupied holes, varies between cones in such a way that θ has beta distribution with parameters α and β .

Let us consider now what could be the best strategy for a bird. First, it is reasonable to assume that the bird is trying to optimize³⁵ the rate of food intake per unit of time. Specifically, a strategy will tell the bird when to leave a cone and start searching the next one. The bird will optimize the rate of food intake, taking into account the average catch at a cone, average time spent on the cone, and the average time of flying to another cone.

Intuitively, if α and β are large, the variability of θ is small: the variance of the beta distribution is $\alpha\beta/(\alpha + \beta)^2(\alpha + \beta + 1)$. In such cases, all cones are about the same: the variability between the cones is due mainly to variability in the binomial distribution with parameters n and $\theta = \alpha/(\alpha + \beta)$. In such cases, there is very little incentive to fly to the next cone before the current one is searched to the end.

However, if α and β are small, in particular, if $\alpha < 1$ and $\beta < 1$, the variability of θ is large. Actually, in the latter case, the distribution of θ has density unbounded at $\theta = 0$ and at $\theta = 1$. This means that most cones will be of two categories only: very rich in prey, when θ is close to 1, and very poor in prey, when θ is close to 0. The optimal strategy is then to assess—as quickly as possible—whether θ is close to 1 or to 0, and to behave accordingly, leaving the cone in the second case.

Now, if after searching k holes the bird found x worms, its assessment of θ is $(\alpha + x)/(\alpha + \beta + k)$ (i.e., equals the Bayes estimate of θ given x successes in k trials). Without going into detail, the optimal strategy specifies, for each k , the threshold for x , below which the bird ought to leave the cone.

³⁵We are using here a convenient terminology based on an analogy with humans. In reality, birds cannot be expected to solve optimization problems, which require computers for humans (see Green, 1980). What we mean here is that in the process of evolution, any adaptation toward a better search strategy is likely to become established, and it is possible that birds use a strategy that is close to optimal.

To determine whether or not the birds follow the optimal strategy is an empirical problem. The experiments involve the use of artificial cones and observation of birds' behavior depending on the findings in the holes searched previously. The preliminary results indicate that birds follow some kind of strategy of breaking or continuing the search of a cone depending on the outcome of the search so far. Whether or not this is an optimal strategy is unclear. But the truly fascinating question here is that if the birds do use a Bayesian strategy, are they are capable of changing the prior distribution? In other words, are birds born with knowledge of a search strategy that is optimal against some fixed α and β that reflect infestation rate of cones over the last hundred years (for example), or can an individual bird change its search strategy in years of higher infestation of its habitat?

Let us investigate briefly the problem of consistency of Bayes estimators. First, let us observe that in the cases of estimators analyzed (of θ in the binomial case with a beta prior, of θ in the Poisson case with a gamma prior, and of μ in the normal distribution with a normal prior), as the sample size increases, the effect of prior distribution decreases to zero. Indeed, if S_n denotes the binomial random variable with probability of success θ , the Bayes estimator of θ for a beta prior satisfies

$$\hat{\theta} = \frac{\alpha + S_n}{\alpha + \beta + n} = \frac{\alpha/n + S_n/n}{(\alpha + \beta)/n + 1} \xrightarrow{P} \theta,$$

since S_n/n converges to θ in probability (and also almost surely).

Similarly, if $X_1, \dots, X_n$ is a random sample from the Poisson distribution with mean θ , and θ has a gamma distribution with parameters α and λ , then the Bayes estimator of θ satisfies

$$\hat{\theta} = \frac{\alpha + \sum_{j=1}^n X_j}{\lambda + n} = \frac{\alpha/n + \bar{X}_n}{\lambda/n + 1} \xrightarrow{P} \theta,$$

by the law of large numbers, which asserts that the sample average $\bar{X}_n$ converges to θ in probability (and also almost surely).

An analogous conclusion for the normal case has already been obtained in Example 16.16. We see therefore that Bayes estimators are consistent; in fact they become increasingly closer to MLE's of the same parameters, regardless of the prior distribution. This property is true for Bayes estimators under some very general conditions, which we will not state here.

One of the problems that a statistician faces quite often is the determination of the sample size: "How big should n be in order that ..." Various conditions may appear in place of the dots. In the case of estimation, these conditions typically state the precision of the estimate, in the sense of the probability of errors of a given size. In case of Bayes estimators, the situation is relatively simple. One of the criteria for determining the sample size may be expressed through the posterior distribution. In the most typical case, one may wish to have a sample size that ensures that the posterior variance is below a certain minimum (of course, such criteria make sense only if the estimator used is unbiased or has a small bias). Thus far we have dealt with point estimation. An estimator was used to produce a single number, hopefully close to an unknown parameter. The natural question arises: What can be said about the distance between the estimator value and the true value of the parameter?

Here the approaches are different, depending on whether or not one can regard the value of θ as a realization of some random variable.

Bayesian Intervals

In the Bayesian scheme, the true value θ is regarded as the observed value of a random variable Ξ with prior distribution (e.g., density) π . Given the vector of observations, $\mathbf{x}$, the posterior density of Ξ is $\pi(\theta|\mathbf{x})$, and we can assess the probability that Ξ lies between two values a and b as

$$P\{a \leq \Xi \leq b | \mathbf{x}\} = \int_a^b \pi(\theta|\mathbf{x}) d\theta. \quad (16.21)$$

■ EXAMPLE 16.21

Suppose that we observe $x = 2$ successes in $n = 3$ Bernoulli trials with unknown probability of success θ , where θ has the prior distribution $\text{BETA}(2, 2)$. Then the posterior distribution of the parameter is again beta, with parameters $\alpha + x = 4$ and $n + \beta - x = 3$. So the posterior density is

$$\frac{\Gamma(7)}{\Gamma(4)\Gamma(3)} \theta^3(1 - \theta)^2 = 60\theta^3 - 120\theta^4 + 60\theta^5, \quad 0 < \theta < 1.$$

The probability that the true value of θ lies below 0.2 equals

$$\int_0^{0.2} (60\theta^3 - 120\theta^4 + 60\theta^5) d\theta = 0.017.$$

The probability that the true value of θ lies above 0.8 equals

$$\int_{0.8}^1 (60\theta^3 - 120\theta^4 + 60\theta^5) d\theta = 0.099,$$

and consequently, the true value of θ lies between 0.2 and 0.8 with probability $1 - 0.017 - 0.099 = 0.884$. The expected value of the posterior distribution is $4/7 = 0.57$, which is the Bayes estimate of θ .

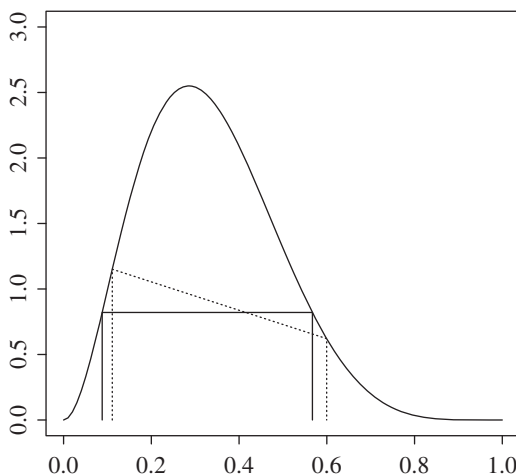


Figure 16.2 The 90% central credible interval (dotted line) and the 90% highest density interval (solid line) for the asymmetric posterior density.

Bayesian intervals are referred to as *credible intervals*. Similarly to the confidence intervals under the frequentist framework (discussed in Chapter 11), one could define many intervals (L, U) such that $P(L \leq \theta \leq U) = 1 - \alpha$. Among them are $(1 - \alpha)100\%$ central credible intervals (L, U) in which $P(\theta < L) = P(\theta > U) = \alpha/2$. Another important subset consists of the *highest density* intervals.

In general, the central credible interval and the highest posterior density (HPD) interval could differ substantially when the posterior density is skewed, as can be seen in Figure 16.2. If the posterior density is unimodal, then the narrowest such interval will be the HPD interval. Regions that contain values with the HPD can be intervals that are symmetric around the mode, but they could also be asymmetric if the posterior distribution is skewed. If the posterior distribution is bimodal, the regions do not even need to be intervals (see Figure 16.2).

Because credible intervals are obtained using posterior probability, it can be said that they contain the parameter of interest with certain (desired) probability. So the interval here is fixed and the parameter is random. Confidence intervals, on the other hand, use probabilities to express how often, in many repetitions, the formula that produces an interval will cover the parameter of interest. In confidence intervals, the interval is random while the parameter is fixed. But unfortunately for the frequentist's approach, once data are collected and the interval is obtained, the probability that it covers the true value of the parameter is either 0 or 1.

■ EXAMPLE 16.22

In Example 16.9, it was shown that in a sample $X_1, \dots, X_n$ from $N(\theta, 1)$, the improper prior $\pi(\theta) = 1$ led to a proper posterior distribution $N(\bar{x}, 1/n)$. The distribution is a symmetric and unimodal, so the $(1 - \alpha)100\%$ HPD interval for θ will be

$$(\bar{x} - z_{\alpha/2}/\sqrt{n}; \bar{x} + z_{\alpha/2}/\sqrt{n}). \quad (16.22)$$

This interval is the same as the $(1 - \alpha)100\%$ confidence interval for θ (discussed in Chapter 11) but it has a different interpretation.

Bayesian Hypotheses Testing

We are testing hypothesis $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, with $\Theta_0 \cup \Theta_1 = \Theta$ and $\Theta_0 \cap \Theta_1 = \emptyset$. In the Bayesian framework, prior distributions $\pi_0(\theta)$ and $\pi_1(\theta)$ are specified for each of the hypotheses, and posterior distributions $\pi_0(\theta|\mathbf{x})$ and $\pi_1(\theta|\mathbf{x})$ can be obtained after collecting observations $\mathbf{x}$.

Definition 16.3.2 The Bayes factor (BF) is defined as the ratio of posterior to prior odds of hypothesis H_0 and posterior to prior odds of H_1 :

$$\text{BF} = \frac{\pi_0(\theta|\mathbf{x})/\pi_0(\theta)}{\pi_1(\theta|\mathbf{x})/\pi_1(\theta)} \quad (16.23)$$

□

Since

$$\pi(\theta_i|\mathbf{x}) = \frac{\pi_i(\mathbf{x}|\theta)\pi_i(\theta)}{\pi_0(\mathbf{x}|\theta)\pi_0(\theta) + \pi_1(\mathbf{x}|\theta)\pi_1(\theta)} \quad (16.24)$$

for $i = 0, 1$, the Bayes factor (16.23) in the case of two simple hypotheses reduces to

$$\text{BF} = \frac{\pi(\mathbf{x}|H_0)}{\pi(\mathbf{x}|H_1)}. \quad (16.25)$$

In the general case of composite hypotheses

$$\text{BF} = \frac{\int_{\Theta_0} \pi_0(\mathbf{x}|\theta) \pi_0(\theta) d\theta}{\int_{\Theta_1} \pi_1(\mathbf{x}|\theta) \pi_1(\theta) d\theta}. \quad (16.26)$$

Interpretation of the Bayes factor is very intuitive; values larger than 1 provide support for the hypothesis H_0 and values less than 1 for hypothesis H_1 .

The following example illustrates use of the Bayes factor for testing simple hypotheses.

■ EXAMPLE 16.23

Let $X_1, \dots, X_{10}$ be observations of Bernoulli trials with the probability of success θ and let the hypotheses to be tested be $H_0 : \theta = \theta_0 = 0.5$ and $H_1 : \theta = \theta_1 = 0.75$. If we assign equal prior odds to H_0 and H_1 then, assuming that there were 6 successes in these 10 trials, the Bayes factor is

$$\begin{aligned} \text{BF} &= \frac{\prod \theta_0^{x_i} (1 - \theta_0)^{1-x_i}}{\prod \theta_1^{x_i} (1 - \theta_1)^{1-x_i}} = \frac{0.5^{\sum x_i} (1 - 0.5)^{10 - \sum x_i}}{0.75^{\sum x_i} (1 - 0.75)^{10 - \sum x_i}} \\ &= \frac{0.5^6 \times 0.5^4}{0.75^6 \times 0.25^4} = 1.4047. \end{aligned} \quad (16.27)$$

The Bayes factor favors H_0 slightly more than H_1 .

The next example considers the case when one hypothesis is simple and the other is composite.

■ EXAMPLE 16.24

Let $X_1, \dots, X_{10}$ be observations of Bernoulli trials with the probability of success θ and hypotheses to be tested be $H_0 : \theta = \theta_0 = 0.5$ and $H_1 : \theta \neq 0.5$. We will assign the uniform prior to values of θ in H_1 , so that $\pi(\theta) = 1$ for $0 < \theta < 1$. Assuming that we also observed 6 successes in 10 trials, the Bayes factor is now

$$\text{BF} = \frac{0.5^{10}}{\int_0^1 \theta^6 (1 - \theta)^4 d\theta}. \quad (16.28)$$

In the denominator of (16.28) we have

$$\int_0^1 \theta^6 (1 - \theta)^4 d\theta = \frac{\Gamma(7)\Gamma(5)}{\Gamma(12)} \int \frac{\Gamma(12)}{\Gamma(7)\Gamma(5)} \theta^6 (1 - \theta)^4 d\theta = \frac{6! 4!}{11!},$$

and finally

$$\text{BF} = \frac{0.5^{10} 11!}{6! 4!} = 2.256.$$

The Bayes factor favors H_0 over H_1 .

■ EXAMPLE 16.25

Continuing Example 16.24, let us now take a prior distribution $\text{BETA}(7, 3)$ which favors values of θ around 0.75. Assuming the same observed data as before (6 successes in 10 Bernoulli trials), we now have

$$\text{BF} = \frac{0.5^{10}}{\int_0^1 252 \theta^6 (1 - \theta)^2 \theta^6 (1 - \theta)^4 d\theta}, \quad (16.29)$$

where

$$252 \theta^6 (1 - \theta)^2 = \frac{\Gamma(10)}{\Gamma(7)\Gamma(3)} \theta^6 (1 - \theta)^2$$

is the density of BETA (7, 3) distribution. In the denominator of (16.29), we now have

$$\int_0^1 252 \theta^6 (1 - \theta)^2 \theta^6 (1 - \theta)^4 d\theta = \frac{252 \Gamma(13) \Gamma(7)}{\Gamma(20)} \int_0^1 \frac{\Gamma(20)}{\Gamma(13) \Gamma(7)} \theta^{12} (1 - \theta)^6 d\theta,$$

and the Bayes factor is

$$\text{BF} = \frac{0.5^{10} 19!}{252 \times 12! 6!} = 1.366$$

and it still favors H_0 over H_1 , but less so, and not enough to make any strong conclusion.

■ EXAMPLE 16.26

Continuing Examples 16.24 and 16.25, let us now assume that there are 60 successes in 100 trials. The relative frequency of success, 0.6, is the same as in the previous examples, but the evidence in the data is stronger since the sample size is now larger. With the same hypotheses to be tested and the same prior as in Example 16.25, the Bayes factor now is

$$\text{BF} = \frac{0.5^{100}}{\int_0^1 252 \theta^6 (1 - \theta)^2 \theta^{60} (1 - \theta)^{40} d\theta} = \frac{0.5^{100} 109!}{252 \times 66! 42!} = 0.591. \quad (16.30)$$

Now the Bayes factor is less than 1; it favors hypothesis H_1 over H_0 .

The readers are encouraged to use the information in the examples above and test the same hypotheses using the classical likelihood ratio tests that were discussed in Chapter 12. The final conclusions in testing under the Bayesian and non-Bayesian framework will be largely consistent, although not always.

A few comments are in order here. First, the Bayes factor can be interpreted as the *weight of evidence* provided by the data. To help with its interpretation, several authors proposed different classification schemes for the value of the Bayes factor (e.g., Kass and Raftery, 1995; Jeffreys, 1961). Second, as Bayes factors update the belief in one hypothesis over the other, they treat both hypotheses equally. Each hypothesis can be accepted or rejected, which is different than in the non-Bayesian framework. In the classical approach to testing hypotheses, the evidence is used only to reject (or not) the so-called null hypothesis; one is not allowed to accept it. Next, the Bayesian factor can be seen as an analog to the likelihood ratio in the frequentist (non-Bayesian) framework as it reduces to likelihood in the case when both hypotheses are simple. And finally, one should be careful when selecting prior distributions for testing hypotheses since the Bayes factor is undefined when prior distributions are improper.

PROBLEMS

- 16.3.1** Let $X_1, \dots, X_n$ be a random sample from a $N(\theta, 1)$ distribution. Using an improper constant prior for parameter θ (see Example 16.9), show that the predictive distribution of the next observation y is $N(\bar{x}, 1 + 1/n)$.
- 16.3.2** A shipment of 100 batteries contains θ defectives. (i) If the prior distribution $\pi(\theta)$ is $\text{BIN}(100, 0.03)$, find the posterior distribution $\pi(\theta|x)$ knowing that one item was randomly selected from the shipment, and was found to not be defective. (ii) Find the predictive probability that the next selected item will also be nondefective.

- 16.3.3** Let $X_1, \dots, X_n$ be a random sample from the $\text{EXP}(\theta)$ distribution, and let the prior distribution of θ be $\text{EXP}(\beta)$. **(i)** Find the Bayes estimator of θ and of $\mu = 1/\theta$ using the squared error loss. **(ii)** Obtain the 90% credible interval for θ and for μ if $n = 6$, $\bar{x} = 4$, and $\beta = 2$.
- 16.3.4** A diastolic blood pressure in a certain population of patients has a $N(\mu, 81)$ distribution. If the blood pressure of 10 randomly selected patients is 76, 71, 82, 63, 64, 75, 81, 78, 66, 62 and μ has a prior distribution $N(70, 1)$, find the predictive distribution of the systolic blood pressure of the next patient randomly selected from the same population.
- 16.3.5** Five people were playing a board game in which to start the game one had to roll a die and obtain either a one or a six dots. It took them respectively 5, 3, 1, 4, and 2 rounds, to start the game. If we assume that the die that they used was fair and that the prior probability θ of getting either one or six dots followed $\text{BETA}(2, 4)$ distribution, find the Bayes estimate of θ assuming the squared error loss.
- 16.3.6** Assume that $X_1, \dots, X_{20}$ is a random sample from the $N(\theta, 1)$. **(i)** Obtain the Bayes factor for testing $H_0: \theta = 0$ against $H_1: \theta \neq 0$ if the prior distribution of θ under H_1 is $N(0.5, 0.25)$. Use $\sum x_i = 7$. **(ii)** Test the same hypothesis in a non-Bayesian framework and compare both results.
- 16.3.7** An insurance company estimates that about 40% of travelers (with the standard deviation of 20%) buy insurance for their airplane tickets. If in a random sample of 100 travelers who purchased an airplane tickets on a given day exactly 35 of them also purchased insurance, find the Bayes estimate of the true proportion of travelers buying insurance assuming the squared error loss (we assume that all travelers in the sample purchased their tickets independently and traveled alone).
- 16.3.8** Let $X, \dots, X_n$ be a random sample from a $\text{GEO}(\theta)$ distribution and let the prior distribution of θ be $\text{BETA}(2, 2)$. **(i)** Find the posterior distribution $\pi(\theta|\mathbf{x})$ and obtain the Bayes estimator of θ assuming squared error loss. **(ii)** Compare the obtained Bayes estimator with the MLE for the same data. **(iii)** Obtain the 95% credible interval for θ .
- 16.3.9** Let $X, \dots, X_{16}$ be a random sample from a $N(\theta, 4)$ distribution and let the prior distribution of θ be $N(0, 0.25)$. **(i)** Find the Bayes estimator of θ assuming squared error loss. **(ii)** Obtain a 95% credible interval for θ .
- 16.3.10** Refer to Example 16.7. Find the Bayes estimator of θ after a was observed under **(i)** Squared error loss. **(ii)** Absolute error loss. **(iii)** Repeat (i) and (ii) assuming that both, a and d were observed.
- 16.3.11** Assume that $X_1, \dots, X_n$ is a random sample from a $\text{BIN}(1, \theta)$ distribution and that the prior distribution of θ is $\text{U}(0, 1)$. Find the Bayes estimator of θ if the loss function is

$$L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2 / [\theta(1 - \theta)]^{3/2}.$$

- 16.3.12** Let $X_1, \dots, X_n$ be a random sample from a $\text{POI}(\theta)$ distribution. **(i)** Using Jeffreys prior, find the posterior distribution for θ . **(ii)** Obtain the 95% credible interval for θ . **(iii)** Obtain the HDP 95% interval for θ if $n = 3$ and the actual observations are 2, 5, and 0. [Hint: The $(1 - \alpha)100\%$ highest density interval is also the shortest $(1 - \alpha)100\%$ credible interval. Use Theorem 8.4.3 and statistical software.]

16.4 FINAL COMMENTS

After years of heated fundamental debates between advocates of Bayesian and non-Bayesian approaches to data analysis, many researchers today have expertise in both and find it helpful to think about analyses from both Bayesian and non-Bayesian perspectives, depending on the problem. We introduced just the main concepts of prior and posterior distributions and provided examples to help understand their properties. We also introduced Bayesian tests of hypotheses and estimation procedures so that the readers could see the analogues to similar methods in non-Bayesian approach. We did not try, however, to introduce Bayesian methodology since there is not enough room in one chapter.

Bayesian inference does not require large sample theory. Asymptotic distributions are not needed since small sample Bayesian inference is carried out the same way as for the large samples. Availability of fast computing and novel computational techniques makes Bayesian analysis useful in most areas of quantitative research. Bayesian methodology allows building multivariate models (also with hierarchical structure) that require powerful computing, for instance for integration, for flexible posterior sampling through iterative simulations within Markov chains. Calculations are carried out using special algorithms, such as Metropolis, Metropolis-Hastings, and Gibbs sampler. Methods of *empirical Bayes* are used for estimating prior distributions from the data. Bayesian data analysis can be performed using readily available software packages such as WinBUGS or R.

Interested readers can find many excellent books and journal articles about Bayesian methodology such as Casella and George (1992), Robert and Casella (1999), Hoff (2009), and Gelman et al. (2013).